

Recursion (All-Questions)

Question # 01

① void printsubsets (int^r arr, int n, int ind, int^r ss)

{
if (ind == n) {

subset [size] = arr[ind];

PS (arr, n, ind+1, subset, size+1);

② void printS (arr, n, ind, size)

{
if (ind == n) {

ss[size] = arr[ind];

PS (arr, n, ind+1, ss, size+1);

③ void PS (arr, n, ind, size)

{
if (ind == n) {

for (int i = 0; i < size; i++)

{ cout << ss[i];

if (i < size-1)

{

cout << " ";

}

}

arr foo (arr, n, ind, size) ✓

{ cout <<

ind == n

return; → to ③

Suppose arr = {1, 2}
ss[0] = {}
ss[1] = {1}
ss[2] = {1, 2}

ind = 1, n = 2, size = 2

ss[1] = {2}
ss = {1, 2}

ind = 2, n = 2, size = 3

1

1

1 1 2

print subse (arr, n, ind+1, subseq, siz);

② PS(→) $n=2, ind=2, siz=2$
 { if (ind == n) ✓
 {
 for (ind →) ✓
 {
 count ++;
 if (→) ✗
 }
 return; // to ① → finish
 ① → ②

PS(arr, n, ind+1, ss, siz);

② PS(→) $n=2, ind=1, siz=0$
 if (→) ✗

subseq[0] = arr[ind]

[0] = [1] ⇒ [0] = 2

subseq[0]
 PS(→)

③ PS(→) $n=2, ind=2, 1$
 if (→) ✓
 { for (→) ✓
 {
 count ++;
 if (→) ✗
 }
 return; to ②

PS(→);

1, 2
1
2

Question # 02

Suppose string is: ABA is $\rightarrow 0 \rightarrow 1$

void permutation(string s, int L, int R)

if (L == R) {

o) for (int i = L; i < R; i++)

{
char t = s[L];
s[L] = s[i];
s[i] = t;
t = i;
s[L] = i;
s[i] = t;
}

a) $\rightarrow$ perm(s, i+1, R);

b) perm(s, L, R)
if (L == R) {

cout << s;
return;

to a)

is

temp = s[L]
s[L] = s[i]
s[i] = temp;
t = i;
s[L] = i;
i = temp;

same again

a) eg for () ✓

char t = s[L]
s[L] = s[i];
s[i] = t;
t = s[L];
s = i;
i = s;

a) $\rightarrow$ perm()

② $P(s, L, R)$

if (—)

call (s)

return go to ②

si

Question # 03

Suppose: arr is: {1, 2}

→ 2 → 2

② $ss(ind^* arr, int n, int sum)$

if (sum == 0) ✓

if (n == 0) ✓

if (arr[n-1] > sum) ✓

return $ss(arr, n-1, sum) \parallel ss(arr, n-1, sum - arr[n-1])$

③ $ss(—) \quad n=1, sum=2$

if (—) ✓

if (n == 0) ✓

if (—) ✓

return $ss(—) \parallel ss(—)$

④ $ss(—) \quad n=0, sum=2$

if (—) ✓

if (n == 0) ✓

return false; → to b

So exist $\parallel$ ⑤ $ss(—) \quad n=0, sum=0$

if (sum == 0) ✓

return true; to b

Question # 04

Suppose : $n = 4$

(a) ind Fib(ind n)

{

if(n=0) x

if(n=1) x

ret fib(n-1) + fib(n-2);

→

(b) fib(-) $n=3$

{ if(-) x

if(-) x

ret fib(n-1) + fib(n-2);

(c) fib(-) $n=2$

if(-) x

if(-) x

ret fib(n-1) + fib(n-2)

(d) fib(-) $n=1$

if(-) x

if(n=1) ✓

ret 1;

(e) fib(-) $n=0$

if(-) ✓

ret 0

→ ret 1+0

(f) fib(-) $n=1$

if(-) x

if(n=1) ✓

ret 1

ret 1+1

② fib(-) $n=2$

if(-) α

if(-) α

ret fib(n-1) + fib(n-2)

① fib(-) $n=1$

if(-) α

if(n=1) $\checkmark$

{ ret 1

② fib(-) $n=0$

if(-)

ret 0

→ ret 1+0

→ ret 1+1

ret 2

Question # 05

suppose: $\text{pow} = n = 2$, $\text{base} = x = 2$

int pow(int x, int n)

{

if(n=0) α

if(n%2=0) $\checkmark$

{ int n = pow(x, n/2);

$x=2, n=1$

② pow(-)

if(-) α

if(-) α

else

ret $x^{\text{pow}(x, n-1)}$;

$\rightarrow 2$
 $\rightarrow 0$
 ③ pow(x, n)
 if (n == 0) ✓
 return 1; to ⑥

return $x^{*1} = 2$; to ①

int h = 2
 return $h * h = 2 * 2 = 4$

Question # 007

suppose : number entered is 123

① int CD(int n) $\rightarrow n = 123$
 {

if (n == 0) x
 return 1 + CD(num/10);

② CD(—) $\rightarrow n = 12$
 if (—) x
 return 1 + CD(num/10);

③ CD(—) $\rightarrow n = 1$
 if (—) x
 return 1 + CD(num/10)

④ CD(—) $\rightarrow n = 0$
 if (n == 0)
 return 0
 return 1 + 0 = 1

return 1 + 1 = 2
 return 1 + 2 = 3

Question # 08

Suppose: arr is {1, 4, 3}

if (int* arr, int n) ⁿ⁼³

if (n == 1) x

int mx = f(arr, n-1)

⑥ f(arr, n-1) n=2

if (n == 1) x

int mx = f(arr, n-1);

⑦ f(arr, n-1) n=1

if (n == 1) ✓

return arr[0];

mx = 1

if (arr[n-1] > mx)

{

return arr[n-1]

}

mx = 3

if (arr[n-1] > mx)

{

return arr[n-1]

[4]

Question # 09

Suppose arr is: {1, 1, 2}

target = 1

n = 3

@int CD(int arr[], int n, int T)

if(n==0) {

return 0;

if(L==T) {

return 1;

if(L < T) {

return 0;

c=0;

if(arr[n-1]==T) {

return c+1;

int c=1;

if(arr[n-1]==T) {

return c+1;

}

return 1; count = 2

(Question # 10)

// Same as reverse string